

- 23** When four identical resistors are connected as shown in diagram 1, the ammeter reads 1.0 A and the voltmeter reads zero.

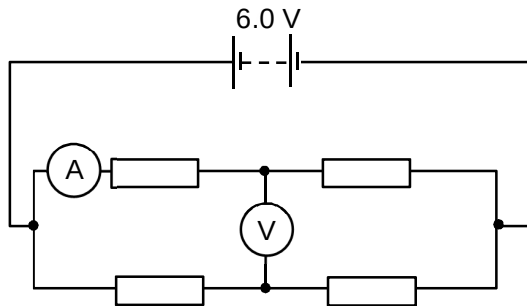


diagram 1

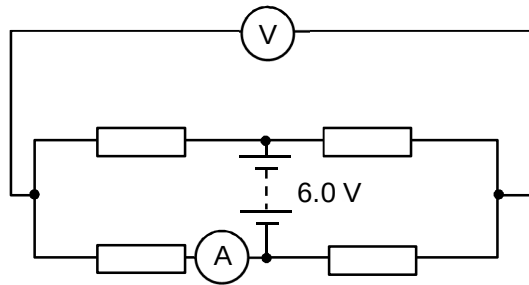


diagram 2

The resistors and meters are reconnected to the supply as shown in diagram 2. What are the meter readings in diagram 2?				
		voltmeter reading / V	ammeter reading / A	
	A	0	1.0	
	B	0	0.5	
	C	3.0	1.0	
	D	6.0	0.5	

Answer: A

Moving the battery source and the meters to the new position does not change the effective layout of the circuit. By circuit analysis the 4 resistors are still connected similar to the first i.e. 2 in series with each other and then these 2 in parallel with the other branch.

Therefore based on the position of the ammeter it is measuring the current in one branch of the circuit hence the current is still 1.0 A and the voltmeter is now measuring across two points in the circuit where the potentials of these two points are the same. So potential difference is 0 V.

OR

Based on diagram 1,

Let the unknown value of the resistors be R.

$$\begin{aligned}
 (1.0)(R+R) &= 6.0 \text{ V} \\
 2R &= 6.0 \text{ V} \\
 R &= 3.0 \, \Omega
 \end{aligned}$$

Based on diagram 2,

$$\begin{aligned}
 I(6.0) &= 6.0 \text{ V} \\
 I &= 1.0 \text{ A}
 \end{aligned}$$

Since the potential difference is measured after the first resistor in both branches the potentials at these two points are the same. Hence potential difference is 0 V.